PV102 Day 6 Afternoon Lab

Balance-of-System Block Diagram

PV102 | CTS Day 22 | Afternoon Lab
Power electronics set system boundaries. | Tuesday, August 11, 2026

Mission

Teams build a labeled block diagram and defend every protection/isolation point.

Safety boundary

This lab is training work. Do not energize or connect equipment outside the instructor-approved procedure. When in doubt, stop, isolate, verify, and ask.

Scenario options

- 12 V training system with PV module, controller, LiFePO4 battery, inverter, DC load, AC load.
- 48 V bus system concept with hybrid inverter/charger, battery bank, PV strings, AC distribution.
- Remote DC-only sensor system with no inverter and limited service visits.

Procedure

- Choose a scenario and place component cards in energy-path order.
- Draw source, storage, conversion, and load boundaries.
- Add conceptual OCPD/disconnect labels where energy could feed a fault or where service isolation is needed.
- Mark manufacturer-instruction-required decisions.
- Run a peer safety check: can someone tell what is energized, isolated, and unknown?
- Handoff the diagram and unresolved decisions.

Acceptance criteria

- Match the real job.
- Show the energy or work path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.
- Give a handoff another crew can use.

Team handoff form

Item	Response
Team members	
Scenario	

Primary decision	
Evidence	
Safety boundary	
Unknowns	
Next safest action	